

**This assignment will not be graded and is only for practice.**

### 3.1 Level 1:

Consider a system of linear equations (System 1):

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$

Answer questions 1 and 2 based on the above data.

1)

If the matrix representation of system (1) is  $Ax = b$ , where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , then

**1 point**

☒  $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$

☐  $b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$

☐  $A = \begin{bmatrix} -2 & -1 & 0 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

☒  $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$ , and  $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Targeted Feedback:

Feedback:

Accepted Answers:

$$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$$

2) System (1) has

**1 point**

[Hint: Solve for  $x_1$ ,  $x_2$ , and  $x_3$ .]

- ☒ a unique solution.
- ☐ no solution.
- ☐ infinitely many solutions.
- ☐ None of the above.

Yes, the answer is correct.

Score: 1

Accepted Answers:

a unique solution.

3) Choose the set of correct options.

**1 point**

[Hint: Think of  $A$  as a  $2 \times 2$  or  $3 \times 3$  matrix, and  $b$  accordingly.]

- ☒ Every system of linear equations has either a unique solution, no solution or infinitely many solutions.
- ☐ If each equation of a system of linear equations is multiplied by a non-zero constant  $c$ , then the solution of the new system of equations is  $c$  times the solution of the old system of equations.
- ☒ If  $Ax = b$  is a system of linear equations which has a solution, then the system of linear equations  $cAx = b$ , where  $c \neq 0$ , will also have a solution.
- ☒ If  $Ax = b$  is a system of linear equations which has a solution, then  $\frac{1}{c}Ax = b$ , where  $c \neq 0$ , will also have a solution.

Yes, the answer is correct.

Score: 1

## Accepted Answers:

Every system of linear equations has either a unique solution, no solution or infinitely many solutions.

If  $Ax = b$  is a system of linear equations which has a solution, then the system of linear equations  $cAx = b$ , where  $c = 0$ , will also have a solution.

If  $Ax = b$  is a system of linear equations which has a solution, then  $\frac{1}{c}Ax = b$ , where  $c \neq 0$ , will also have a solution.

4) The Plane 1 and Plane 2 in Figure M2W1AQ3, correspond to two different linear equations, **1 point**  
which form a system of linear equations.

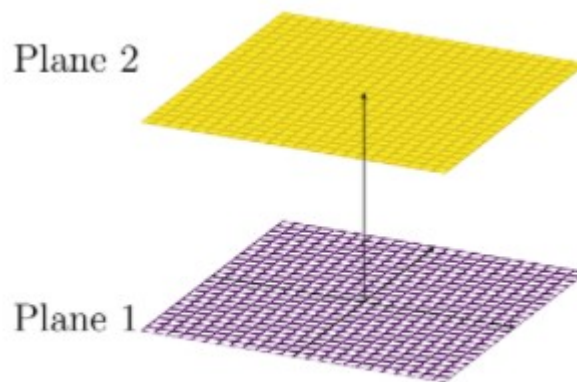


Figure M2W1AQ3

The above system of linear equations has

[Hint: If a system of linear equations has a solution, then the point corresponding to the solution must lie on each plane corresponding to each linear equation of the given system.]

- ☐ a unique solution.
- ☒ no solution.
- ☐ infinitely many solutions.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

## Accepted Answers:

no solution.

5) Consider the geometric representations (Figures (a), (b), and (c)) of three systems of linear equations. **1 point**

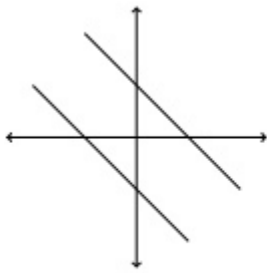


Figure (a)

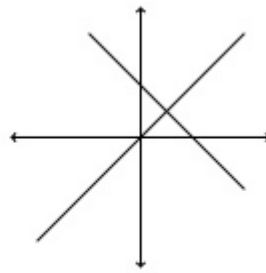


Figure (b)

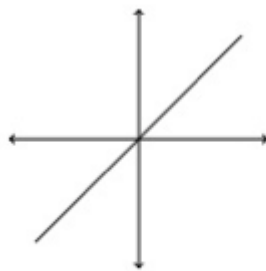


Figure (c)

Choose the set of correct options.

- ☒ Figure (a) represents a system of linear equations which has no solution.
- ☐ Figure (a) represents a system of linear equations which has infinitely many solutions.
- ☒ Figure (b) represents a system of linear equations which has a unique solution.
- ☐ Figure (b) represents a system of linear equations which has infinitely many solutions.
- ☒ Figure (c) represents a system of linear equations which has infinitely many solutions.
- ☐ Figure (c) represents a system of linear equations which has no solution.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Figure (a) represents a system of linear equations which has no solution.

Figure (b) represents a system of linear equations which has a unique solution.

Figure (c) represents a system of linear equations which has infinitely many solutions.

**3.2 Level 2:**

6)  $2x_1 + 3x_2 = 6$  **1 point**

Consider a system of equations:  $-2x_1 + kx_2 = d$  Choose the set of correct options.

$$4x_1 + 6x_2 = 12$$

- ☐  $Ax = b$  represents the above system, where  $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ ,  $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$ , and  $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$
- ☐ The system has no solution if  $k = -3, d = 0$ .
- ☐ The system has a unique solution if  $k = 3, d = 0$ .
- ☐ The system has infinitely many solutions if  $k = -3, d = 6$ .
- ☐ The system has infinitely many solutions if  $k = -3, d = -6$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$Ax = b$  represents the above system, where  $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ ,  $A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$ , and  $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$

The system has no solution if  $k = -3, d = 0$ .

The system has a unique solution if  $k = 3, d = 0$ .

The system has infinitely many solutions if  $k = -3, d = -6$ .

7) Let  $x_1$  and  $x_2$  be solutions of the system of linear equations  $Ax = b$ . Which of the following options are correct? **1 point**

- ☐  $x_1 + x_2$  is a solution of the system of linear equations  $Ax = b$ .
- ☐  $x_1 + x_2$  is a solution of the system of linear equations  $Ax = 2b$ .
- ☐  $x_1 - x_2$  is a solution of the system of linear equations  $Ax = b$ .
- ☐  $x_1 - x_2$  is a solution of the system of linear equations  $Ax = 0$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1 + x_2$  is a solution of the system of linear equations  $Ax = 2b$ .

$x_1 - x_2$  is a solution of the system of linear equations  $Ax = 0$ .

8) Let  $v$  be a solution of the systems of linear equations  $A_1x = b$  and  $A_2x = b$ . Which of the following options are correct ?

**1 point**

- ☐  $v$  is a solution of the system of linear equations  $(A_1 + A_2)x = b$ .
- ☐  $v$  is a solution of the system of linear equations  $(A_1 + A_2)x = 2b$ .
- ☐  $v$  is a solution of the system of linear equations  $(A_1 - A_2)x = 0$ .
- ☐  $v$  is a solution of the system of linear equations  $(A_1 - A_2)x = b$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$v$  is a solution of the system of linear equations  $(A_1 + A_2)x = 2b$ .

$v$  is a solution of the system of linear equations  $(A_1 - A_2)x = 0$ .

9) Consider a system of equations: 
$$\begin{matrix} x_1 - 3x_2 = 4 \\ 3x_1 + kx_2 = -12 \end{matrix}$$
 Where  $k \in \mathbb{R}$ . If the given system has a unique solution, then  $k$  should not be equal to

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -9

**1 point**

10) Consider two system of linear equations:  
System 1:

$$x_1 + 2x_2 = -5$$

$$0x_1 - x_2 = 5$$

System 2:

$$-2x_3 + x_4 = -5$$

$$3x_3 + x_4 = 5$$

Suppose there is another system of linear equation given by

$$(1 - 2)(x_1 + x_3) + (2 + 1)(x_2 + x_4) = m$$

$$(0 + 3)(x_1 + x_3) + (-1 + 1)(x_2 + x_4) = n$$

for some real values of  $m$  and  $n$ . Find the value of  $n - m$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 46

**1 point**

Your score is: 4/10